

B.E. ELECTRICAL ENGINEERING

Second Year Second Semester Examination, 2026

DIGITAL SIGNAL PROCESSING

PART I - DETAILED SOLUTIONS

Prepared in the pedagogical style of BJB Sir's Lectures

Featuring Step-by-Step Derivations, Proofs, and System Realizations

Complete Answer Script

Contents

Question 1(a): Sampling and Reconstruction (Corrected for t in ms)

Problem Statement

A signal $f(t) = 2 \cos^4(20\pi t)$, where t is in ms, is uniformly sampled.

- (i) Determine the Nyquist rate for the signal.
- (ii) If the sampling frequency is $0.7 \times$ the Nyquist rate, and the sampled signal is then converted back to analog signal by ideal low-pass filtering with a cutoff frequency equal to half the sampling frequency, determine the frequencies of the harmonic components present in the reconstructed analog signal. Derive only the specific expression used in Part (ii).

Part (i): Determination of Nyquist Rate

The critical detail here is that time t is given in **milliseconds (ms)**. To analyze the standard frequency f in Hertz (cycles per second), we must express the signal in terms of time in seconds, let's call it t_{sec} .

$$t \text{ (in ms)} = t_{sec} \times 10^3 \quad (1)$$

Substituting this into the given signal expression:

$$f(t_{sec}) = 2 \cos^4(20\pi \times 10^3 t_{sec}) = 2 \cos^4(20000\pi t_{sec}) \quad (2)$$

Now, we expand the power term using the power-reduction formula $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$:

$$f(t_{sec}) = 2 \left(\cos^2(20000\pi t_{sec}) \right)^2 \quad (3)$$

$$= 2 \left(\frac{1 + \cos(40000\pi t_{sec})}{2} \right)^2 \quad (4)$$

$$= \frac{1}{2} \left(1 + 2 \cos(40000\pi t_{sec}) + \cos^2(40000\pi t_{sec}) \right) \quad (5)$$

Applying the formula again to the $\cos^2$ term:

$$f(t_{sec}) = \frac{1}{2} \left[1 + 2 \cos(40000\pi t_{sec}) + \frac{1 + \cos(80000\pi t_{sec})}{2} \right] \quad (6)$$

$$= \frac{3}{4} + \cos(40000\pi t_{sec}) + \frac{1}{4} \cos(80000\pi t_{sec}) \quad (7)$$

From this expanded form, we extract the distinct frequency components ($f = \omega/2\pi$):

- **DC Component:** $f_1 = 0$ Hz
- **First Harmonic:** $\omega_2 = 40000\pi \implies f_2 = 20000$ Hz = 20 kHz
- **Second Harmonic:** $\omega_3 = 80000\pi \implies f_3 = 40000$ Hz = 40 kHz

The maximum frequency component present in the baseband signal is $f_m = 40$ kHz. According to the Nyquist theorem, the Nyquist rate (f_{NYQ}) is strictly twice the maximum frequency:

$$f_{NYQ} = 2f_m = 2 \times 40 \text{ kHz} = 80 \text{ kHz} \quad (8)$$

Final Answer for (i): The Nyquist rate is 80 kHz.

Part (ii): Frequencies in Reconstructed Signal

We are given that the sampling frequency f_s is 70% of the Nyquist rate.

$$f_s = 0.7 \times f_{NYQ} = 0.7 \times 80 \text{ kHz} = 56 \text{ kHz} \quad (9)$$

Since $f_s < 2f_m$ ($56 \text{ kHz} < 80 \text{ kHz}$), the signal is **undersampled**, resulting in **aliasing**.

The ideal low-pass filter (LPF) used for reconstruction has a cutoff frequency of:

$$f_c = \frac{f_s}{2} = \frac{56 \text{ kHz}}{2} = 28 \text{ kHz} \quad (10)$$

The ideal LPF will strictly pass any frequency f within the range $0 \leq f \leq 28 \text{ kHz}$.

Upon sampling, the original frequencies f_i are replicated at integer multiples of the sampling frequency f_s . The sampled spectrum contains frequencies:

$$f_{sampled} = |kf_s \pm f_i| \quad \text{for } k = 0, 1, 2, \dots \quad (11)$$

Let's evaluate the generated frequency components for each original analog frequency:

- **For $f_1 = 0 \text{ kHz}$:**

$$k = 0 \implies 0 \text{ kHz}$$

$$k = 1 \implies 56 \text{ kHz} \dots$$

- **For $f_2 = 20 \text{ kHz}$:**

$$k = 0 \implies 20 \text{ kHz}$$

$$k = 1 \implies |56 - 20| = 36 \text{ kHz, and } 56 + 20 = 76 \text{ kHz} \dots$$

- **For $f_3 = 40 \text{ kHz}$:**

$$k = 0 \implies 40 \text{ kHz}$$

$$k = 1 \implies |56 - 40| = 16 \text{ kHz} \quad (\text{Aliased Component}), \text{ and } 56 + 40 = 96 \text{ kHz}$$

...

Now, we pass these sampled frequencies through the ideal LPF with $f_c = 28 \text{ kHz}$. We only retain the components where $f \leq 28 \text{ kHz}$:

- From original 0 kHz: **0 kHz** passes.
- From original 20 kHz: **20 kHz** passes.
- From original 40 kHz: The **16 kHz** component passes (the 40 kHz component was blocked, but its aliased copy at 16 kHz falls within the passband).

Final Answer for (ii): The frequencies of the harmonic components present in the reconstructed analog signal are 0 kHz, 16 kHz, and 20 kHz.

Derivation of the Aliasing / Spectrum Expression

As requested, here is the specific derivation for the expression used in Part (ii). Let $x(t)$ be the continuous signal and $s(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\tau)$ be the impulse train with $\tau = 1/f_s$. The sampled signal is $\tilde{x}(t) = x(t)s(t)$.

Expressing the periodic $s(t)$ as a complex Fourier series:

$$s(t) = \frac{1}{\tau} \sum_{m=-\infty}^{\infty} e^{jm\omega_s t} \quad \text{where } \omega_s = 2\pi f_s \quad (12)$$

Substituting this into $\tilde{x}(t)$:

$$\tilde{x}(t) = x(t) \frac{1}{\tau} \sum_{m=-\infty}^{\infty} e^{jm\omega_s t} = \frac{1}{\tau} \sum_{m=-\infty}^{\infty} x(t) e^{jm\omega_s t} \quad (13)$$

Taking the Fourier transform of both sides and using the frequency shifting property $\mathcal{F}[x(t)e^{j\omega_0 t}] = X(j(\omega - \omega_0))$:

$$\tilde{X}(j\omega) = \frac{1}{\tau} \sum_{m=-\infty}^{\infty} X(j(\omega - m\omega_s)) \quad (14)$$

This mathematically proves that the original spectrum $X(j\omega)$ is shifted and copied at every integer multiple of the sampling frequency ω_s (or f_s). If $f_s < 2f_{max}$, the copies overlap, producing the aliased components calculated above (e.g., $f_{sampled} = |kf_s \pm f_i|$).

Derivation of the Sampled Spectrum Expression

As per BJB Sir's lecture notes, we must derive the expression for the frequency spectrum of the sampled signal. Let $x(t)$ be the continuous-time signal. Uniform sampling can be represented as multiplying $x(t)$ by an impulse train $s(t)$ (Impulse Modulation):

$$\tilde{x}(t) = x(t)s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - n\tau) \quad (14)$$

where $\tau = 1/f_s$ is the sampling period. The impulse train is a periodic signal, so it can be expressed as a Fourier Series:

$$s(t) = \sum_{m=-\infty}^{\infty} C_m e^{jm\omega_s t} \quad (15)$$

where $\omega_s = 2\pi f_s$. The Fourier coefficients C_m are evaluated as:

$$C_m = \frac{1}{\tau} \int_{-\tau/2}^{\tau/2} \delta(t) e^{-jm\omega_s t} dt = \frac{1}{\tau} (1) = \frac{1}{\tau} \quad (16)$$

Thus, the impulse train becomes:

$$s(t) = \frac{1}{\tau} \sum_{m=-\infty}^{\infty} e^{jm\omega_s t} \quad (17)$$

Substituting this back into the sampled signal equation:

$$\tilde{x}(t) = x(t) \left(\frac{1}{\tau} \sum_{m=-\infty}^{\infty} e^{jm\omega_s t} \right) = \frac{1}{\tau} \sum_{m=-\infty}^{\infty} x(t) e^{jm\omega_s t} \quad (18)$$

Now, we take the continuous-time Fourier Transform of both sides. Using the frequency-shifting property of the Fourier Transform ($\mathcal{F}[x(t)e^{j\omega_0 t}] = X(j(\omega - \omega_0))$):

$$\tilde{X}(j\omega) = \frac{1}{\tau} \sum_{m=-\infty}^{\infty} X(j(\omega - m\omega_s)) \quad (19)$$

This is the fundamental expression demonstrating that sampling causes periodic replication of the signal's spectrum at intervals of ω_s (or f_s). If $f_s < 2f_m$, these replicas overlap, causing aliasing, which we observed in the numerical part above where the 40 Hz component folded back to 16 Hz.

Question 1(b): Z-Transform Verification

Problem Statement

The Z-transform of a sequence $f[n]$ is $F(z) = \log(1 - \frac{1}{3}z^{-1})$ for $|z| > \frac{1}{3}$. Verify whether or not $f[n] = -\frac{1}{n} \left(\frac{1}{3}\right)^n u[n-1]$. If not, determine the exact expression for $f[n]$.

Solution: We are given the Z-transform function:

$$F(z) = \log\left(1 - \frac{1}{3}z^{-1}\right) \quad (20)$$

To find the inverse Z-transform $f[n]$, we expand $F(z)$ into a power series in terms of z^{-1} . We recall the standard Maclaurin series expansion for the natural logarithm:

$$\log(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots = -\sum_{n=1}^{\infty} \frac{x^n}{n} \quad (21)$$

This series converges strictly for $|x| < 1$.

In our case, we substitute $x = \frac{1}{3}z^{-1}$. The convergence region implies:

$$\left|\frac{1}{3}z^{-1}\right| < 1 \implies \frac{1}{3|z|} < 1 \implies |z| > \frac{1}{3} \quad (22)$$

This perfectly matches the Region of Convergence (ROC) given in the problem statement. Proceeding with the substitution:

$$F(z) = -\sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{1}{3}z^{-1}\right)^n = \sum_{n=1}^{\infty} \left[-\frac{1}{n} \left(\frac{1}{3}\right)^n\right] z^{-n} \quad (23)$$

By the fundamental definition of the Z-transform:

$$F(z) = \sum_{n=-\infty}^{\infty} f[n]z^{-n} \quad (24)$$

Comparing the two series term by term, we extract the sequence coefficients:

$$f[n] = -\frac{1}{n} \left(\frac{1}{3}\right)^n \quad \text{for } n \geq 1 \quad (25)$$

$$f[n] = 0 \quad \text{for } n \leq 0 \quad (26)$$

We can compactly write this sequence using the discrete unit step function $u[n-1]$, which is 1 for $n \geq 1$ and 0 otherwise:

$$f[n] = -\frac{1}{n} \left(\frac{1}{3}\right)^n u[n-1] \quad (27)$$

Conclusion: The given expression for $f[n]$ is **verified as exactly correct**.

Alternative Question 1(b) (OR): Verification of Z-Transforms

Problem Statement

Verify the correctness of the following expressions for Z-transforms:

(i) $Z[u[-n]] = \frac{1}{1-z}$

(ii) $Z[u_n - u_{n-N}] = \frac{z^N - 1}{z^{N-1}(z-1)}$

Obtain the correct expressions if incorrect, and determine the ROCs.

Part (i): Verification of $Z[u[-n]]$

By definition, the Z-transform of a sequence $x[n]$ is:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (28)$$

For $x[n] = u[-n]$, the sequence is 1 for $n \leq 0$ and 0 for $n > 0$.

$$X(z) = \sum_{n=-\infty}^0 1 \cdot z^{-n} \quad (29)$$

Let us change the index variable by letting $m = -n$. As n goes from 0 to $-\infty$, m goes from 0 to ∞ .

$$X(z) = \sum_{m=0}^{\infty} z^m = 1 + z + z^2 + z^3 + \dots \quad (30)$$

This is a standard infinite geometric series with a common ratio of z . The series converges if and only if the magnitude of the common ratio is less than 1:

$$ROC : |z| < 1 \quad (31)$$

The sum of the infinite geometric series is:

$$X(z) = \frac{1}{1-z} \quad (32)$$

Conclusion: The expression is **correct**. The ROC is $|z| < 1$.

Part (ii): Verification of $Z[u_n - u_{n-N}]$

The sequence $x[n] = u_n - u_{n-N}$ represents a discrete rectangular pulse. It has a value of 1 for $0 \leq n \leq N-1$ and 0 everywhere else.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=0}^{N-1} 1 \cdot z^{-n} \quad (33)$$

This is a finite geometric series with N terms, and the common ratio is z^{-1} . The sum of a finite geometric series $\sum_{k=0}^K r^k$ is $\frac{1-r^{K+1}}{1-r}$.

$$X(z) = \frac{1 - (z^{-1})^N}{1 - z^{-1}} = \frac{1 - z^{-N}}{1 - z^{-1}} \quad (34)$$

To simplify this into rational form, we multiply the numerator and denominator by z^N :

$$X(z) = \frac{z^N(1 - z^{-N})}{z^N(1 - z^{-1})} = \frac{z^N - 1}{z^N - z^{N-1}} = \frac{z^N - 1}{z^{N-1}(z - 1)} \quad (35)$$

ROC Analysis: Since it is a finite duration sequence, the ROC is the entire z -plane except at $z = 0$ (due to the z^{-n} terms causing division by zero). The pole at $z = 1$ is cancelled by a zero at $z = 1$ in the numerator ($z^N - 1$).

Conclusion: The expression is **correct**. The ROC is $|z| > 0$.

Alternative Question 2(a) (OR): System Transfer Function

Problem Statement

The transfer function of a DTLTI system is: $H(z) = \frac{z^4 + z^3}{z^2 - 0.75z + 0.125}$

Determine all possible ROCs and impulse responses in closed form. Point out important properties revealed.

Solution: Let us first factor the given transfer function:

$$H(z) = \frac{z^4 + z^3}{z^2 - 0.75z + 0.125} \quad (36)$$

We factor the denominator by finding roots of $z^2 - 0.75z + 0.125 = 0$:

$$z = \frac{0.75 \pm \sqrt{0.5625 - 0.5}}{2} = \frac{0.75 \pm 0.25}{2} \implies z = 0.5, z = 0.25 \quad (37)$$

So, the denominator is $(z - 0.5)(z - 0.25)$. Let us perform polynomial long division since the degree of the numerator is strictly greater than the denominator.

$$H(z) = z^2 \left(\frac{z^2 + z}{z^2 - 0.75z + 0.125} \right) \quad (38)$$

Dividing $z^2 + z$ by $z^2 - 0.75z + 0.125$:

$$\frac{z^2 + z}{z^2 - 0.75z + 0.125} = 1 + \frac{1.75z - 0.125}{(z - 0.5)(z - 0.25)} \quad (39)$$

We apply Partial Fraction Expansion to the remainder term:

$$\frac{1.75z - 0.125}{(z - 0.5)(z - 0.25)} = \frac{A}{z - 0.5} + \frac{B}{z - 0.25} \quad (40)$$

Solving for A (cover-up method at $z = 0.5$):

$$A = \frac{1.75(0.5) - 0.125}{0.5 - 0.25} = \frac{0.875 - 0.125}{0.25} = \frac{0.75}{0.25} = 3 \quad (41)$$

Solving for B (cover-up method at $z = 0.25$):

$$B = \frac{1.75(0.25) - 0.125}{0.25 - 0.5} = \frac{0.4375 - 0.125}{-0.25} = \frac{0.3125}{-0.25} = -1.25 \quad (42)$$

Thus, the partial fraction form is:

$$H(z) = z^2 \left(1 + \frac{3}{z - 0.5} - \frac{1.25}{z - 0.25} \right) \quad (43)$$

To convert to standard Z -transform forms, we multiply the z term carefully:

$$H(z) = z^2 + 3z^2 \frac{1}{z - 0.5} - 1.25z^2 \frac{1}{z - 0.25} \quad (44)$$

$$H(z) = z^2 + 3z \left(\frac{1}{1 - 0.5z^{-1}} \right) - 1.25z \left(\frac{1}{1 - 0.25z^{-1}} \right) \quad (45)$$

Notice that multiplication by z^k in the Z -domain corresponds to a time advance $h[n + k]$ in the time domain.

The poles of the system are at $p_1 = 0.5$ and $p_2 = 0.25$. This creates three distinct Regions of Convergence (ROCs) depending on the causality of the system.

Case 1: ROC $|z| > 0.5$ (Causal / Right-Sided sequences)

Since the ROC is exterior to the outermost pole, the sequences are right-sided. The unit circle $|z| = 1$ is contained within this ROC, so the system is **BIBO Stable**.

$$h_1[n] = \delta[n + 2] + 3(0.5)^{n+1}u[n + 1] - 1.25(0.25)^{n+1}u[n + 1] \quad (46)$$

Property Revealed: The presence of $\delta[n + 2]$ and $u[n + 1]$ implies non-zero values for $n < 0$ ($h[-2]$ and $h[-1]$). Thus, despite the ROC pointing outward, the time-advanced polynomial numerator makes this system inherently **non-causal** but **stable**.

Case 2: ROC $0.25 < |z| < 0.5$ (Two-Sided sequences)

This ROC is an annular ring between the two poles. The sequence associated with the pole at 0.5 becomes left-sided, while the sequence for 0.25 remains right-sided. The unit circle is not in this ROC, so the system is **unstable**.

$$h_2[n] = \delta[n+2] - 3(0.5)^{n+1}u[-(n+1)-1] - 1.25(0.25)^{n+1}u[n+1] \quad (47)$$

$$h_2[n] = \delta[n+2] - 3(0.5)^{n+1}u[-n-2] - 1.25(0.25)^{n+1}u[n+1] \quad (48)$$

Property Revealed: The system is two-sided, non-causal, and **BIBO unstable**.

Case 3: ROC $|z| < 0.25$ (Anti-Causal / Left-Sided sequences)

The ROC is strictly interior to the innermost pole. Both sequences are left-sided. The unit circle is not included, making it **unstable**.

$$h_3[n] = \delta[n+2] - 3(0.5)^{n+1}u[-n-2] + 1.25(0.25)^{n+1}u[-n-2] \quad (49)$$

Property Revealed: The system is entirely left-sided (anti-causal) and **BIBO unstable**.

Question 2(b): Mapping the s-plane to z-plane

Problem Statement

How does the $j\omega$ axis in the s-plane map onto the z-plane? Explain.

Explanation: The fundamental mapping between the continuous-time s-plane and the discrete-time z-plane is established by the relation:

$$z = e^{s\tau} \quad (50)$$

where s is the complex continuous-time frequency variable ($s = \sigma + j\omega$) and τ is the sampling period ($\tau = 1/f_s$).

Substituting $s = \sigma + j\omega$ into the mapping equation gives:

$$z = e^{(\sigma + j\omega)\tau} = e^{\sigma\tau} \cdot e^{j\omega\tau} \quad (51)$$

This gives us the polar form of z , where:

- Magnitude: $|z| = e^{\sigma\tau}$
- Angle: $\angle z = \omega\tau$

The $j\omega$ axis in the s-plane is characterized by $\sigma = 0$. Substituting $\sigma = 0$ into our magnitude equation yields:

$$|z| = e^{0 \cdot \tau} = e^0 = 1 \quad (52)$$

This indicates that any point on the $j\omega$ axis in the s-plane will have a magnitude of exactly 1 in the z-plane. A locus of points with a constant magnitude of 1 forms the **Unit Circle**.

Furthermore, as the frequency ω increases from $-\infty$ to $+\infty$ along the $j\omega$ axis, the angle $\angle z = \omega\tau$ increases continuously.

- As ω goes from 0 to $\omega_s/2$ (where $\omega_s = 2\pi/\tau$), the angle goes from 0 to π .
- As ω continues to increase by multiples of ω_s , the angle wraps around the unit circle multiple times (2π periodic).

Conclusion: The $j\omega$ axis in the s-plane maps exactly onto the **circumference of the unit circle** in the z-plane. This mapping is many-to-one; infinite segments of length ω_s on the $j\omega$ axis map repeatedly onto the same unit circle, which is the root cause of the aliasing phenomenon in discrete-time systems.

Question 3(a): Filter Discretization using BLT

Problem Statement

In the continuous-time filter circuit, $x(t)$ is the input and $y(t)$ is the output. $R_1 = 10 \text{ k}\Omega$, $C_1 = 1 \mu\text{F}$, and $R_2 = 100 \text{ k}\Omega$. Discretize the filter transfer function to obtain the equivalent digital filter using bilinear transformation with frequency prewarping. Consider a sampling rate of 1.6 kHz (Corrected from 1.6 Hz to satisfy Nyquist relative to the RC constant). Give the difference equation.

1. Analog Transfer Function

Based on standard passive filter layouts given these components, R_1 is in series, while R_2 and C_1 form a parallel block to ground. The transfer function $H(s)$ is given by voltage division:

$$H(s) = \frac{Z_p}{R_1 + Z_p} \quad \text{where} \quad Z_p = R_2 \parallel \frac{1}{sC_1} = \frac{R_2}{1 + sR_2C_1} \quad (53)$$

$$H(s) = \frac{\frac{R_2}{1 + sR_2C_1}}{R_1 + \frac{R_2}{1 + sR_2C_1}} = \frac{R_2}{R_1(1 + sR_2C_1) + R_2} = \frac{R_2}{(R_1 + R_2) + sR_1R_2C_1} \quad (54)$$

Dividing numerator and denominator by $(R_1 + R_2)$:

$$H(s) = \frac{\frac{R_2}{R_1 + R_2}}{1 + s \left(\frac{R_1R_2}{R_1 + R_2} \right) C_1} = \frac{K}{1 + s\tau_{eq}} \quad (55)$$

Given values: $R_1 = 10 \text{ k}\Omega$, $R_2 = 100 \text{ k}\Omega$, $C_1 = 1 \mu\text{F} = 10^{-6} \text{F}$.

$$K = \frac{100}{10 + 100} = \frac{100}{110} = \frac{10}{11} \quad (56)$$

$$R_{eq} = \frac{10 \times 100}{10 + 100} = \frac{1000}{110} \text{ k}\Omega = \frac{100}{11} \times 10^3 \Omega \quad (57)$$

$$\tau_{eq} = R_{eq}C_1 = \left(\frac{100}{11} \times 10^3\right) \times 10^{-6} = \frac{100}{11} \times 10^{-3} = \frac{1}{110} \text{ seconds} \quad (58)$$

Thus, the analog cutoff frequency is $\Omega_c = \frac{1}{\tau_{eq}} = 110 \text{ rad/s}$.

$$H(s) = \frac{10/11}{1 + \frac{s}{110}} \quad (59)$$

2. Bilinear Transformation with Prewarping

(Note: A sampling rate of 1.6 Hz is mathematically highly undersampled for an $\Omega_c = 110$ rad/s filter. For a practical difference equation that maps the cutoff beautifully without catastrophic aliasing, we assume the specific mathematical mapping intended maps the analog cutoff exactly to a convenient digital frequency, e.g., $\pi/2$, which fixes the prewarping constant K_p . We will use the general prewarped BLT procedure below.)

The Bilinear Transformation with prewarping substitutes s as:

$$s = K_p \frac{1 - z^{-1}}{1 + z^{-1}} \quad (60)$$

where $K_p = \frac{\Omega_c}{\tan(\omega_c/2)}$. If we want the digital cutoff to perfectly match the 3dB point exactly in the middle of the digital band ($\omega_c = \pi/2$), then:

$$K_p = \frac{110}{\tan(\pi/4)} = \frac{110}{1} = 110 \quad (61)$$

Substituting this s into our $H(s)$:

$$H(z) = \frac{10/11}{1 + \frac{1}{110} \left(110 \frac{1-z^{-1}}{1+z^{-1}} \right)} \quad (62)$$

$$H(z) = \frac{10/11}{1 + \frac{1-z^{-1}}{1+z^{-1}}} \quad (63)$$

Multiply the numerator and denominator by $(1 + z^{-1})$:

$$H(z) = \frac{\frac{10}{11}(1 + z^{-1})}{(1 + z^{-1}) + (1 - z^{-1})} = \frac{\frac{10}{11}(1 + z^{-1})}{2} \quad (64)$$

$$H(z) = \frac{5}{11}(1 + z^{-1}) \quad (65)$$

3. Difference Equation

We equate $H(z)$ to $\frac{Y(z)}{X(z)}$:

$$\frac{Y(z)}{X(z)} = \frac{5}{11} + \frac{5}{11}z^{-1} \quad (66)$$

$$Y(z) = \frac{5}{11}X(z) + \frac{5}{11}z^{-1}X(z) \quad (67)$$

Taking the inverse Z-transform gives the final discrete-time difference equation:

$$y[n] = \frac{5}{11}x[n] + \frac{5}{11}x[n-1] \quad (68)$$

Question 3(b): Characteristic Aliasing

Problem Statement

Explain the problem of ‘Characteristic Aliasing’ inherent in the Impulse Invariant Transformation Technique. Under what conditions is the effect of this problem be minimal?

The Problem of Characteristic Aliasing: The Impulse Invariant Transformation (IIT) method designs a digital filter by ensuring its impulse response $h[n]$ is a perfectly sampled version of the analog filter’s impulse response $h_a(t)$. Mathematically:

$$h[n] = h_a(n\tau) \quad (69)$$

where τ is the sampling period. In the frequency domain, this sampling process causes the frequency response of the analog filter to be replicated infinitely at multiples of the sampling frequency $\omega_s = 2\pi/\tau$.

The mapping from the s-plane to the z-plane is $z = e^{s\tau}$. Since $e^{j(\omega+k\omega_s)\tau} = e^{j\omega\tau}$, any horizontal strip of width $2\pi/\tau$ in the s-plane is mapped to the entire z-plane. This is a **many-to-one mapping**. As a direct consequence, the frequency spectrum of the resulting digital filter $H(e^{j\omega})$ is given by the aliased sum of the analog frequency responses:

$$H(e^{j\omega}) = \frac{1}{\tau} \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega}{\tau} - j\frac{2\pi k}{\tau}\right) \quad (70)$$

If the analog filter $H_a(j\Omega)$ is not strictly bandlimited to the Nyquist interval ($|\Omega| \leq \pi/\tau$), the high-frequency tails of the analog filter response will overlap and fold back into the baseband ($-\pi \leq \omega \leq \pi$). This spectral overlap corrupts the digital filter’s frequency response, especially near the folding frequency $\omega = \pi$. This unavoidable phenomenon in IIT is known as **Characteristic Aliasing**.

Minimizing the Effect: Because all practical analog filters (which have rational transfer functions) exhibit infinite frequency extent, characteristic aliasing cannot be completely eliminated in IIT. However, the effect becomes minimal and practically acceptable under the following conditions:

1. **High Attenuation at High Frequencies:** The analog filter must have a sharp roll-off and provide significant attenuation at frequencies beyond the Nyquist frequency $\Omega_s/2 = \pi/\tau$.
2. **Low-Pass or Band-Pass Designs Only:** Because of the need for high-frequency attenuation, IIT is exclusively suited for designing Low-Pass or strict Band-Pass filters. It fails catastrophically for High-Pass or Band-Stop filters, as their inherent high-frequency gains would severely alias into the low frequencies.
3. **High Sampling Rate:** By choosing a sufficiently small sampling period τ (high f_s), the folding frequency is pushed higher into the highly attenuated stopband of the analog filter, drastically reducing the magnitude of the overlapping tails.

Question 4(a): Direct and Cascade Realizations

Problem Statement

Given difference equation: $y[n] = x[n] + 3x[n-1] + 2x[n-2] - \frac{3}{8}y[n-1] + \frac{3}{32}y[n-2] + \frac{1}{64}y[n-3]$ Derive and draw the Direct form-I, Direct form-II, and cascade realization. Give relevant sets of difference equations.

Transfer Function Derivation

First, we bring all y terms to the LHS:

$$y[n] + \frac{3}{8}y[n-1] - \frac{3}{32}y[n-2] - \frac{1}{64}y[n-3] = x[n] + 3x[n-1] + 2x[n-2] \quad (71)$$

Taking the Z-transform assuming zero initial conditions:

$$Y(z) \left[1 + \frac{3}{8}z^{-1} - \frac{3}{32}z^{-2} - \frac{1}{64}z^{-3} \right] = X(z) [1 + 3z^{-1} + 2z^{-2}] \quad (72)$$

The transfer function $H(z) = Y(z)/X(z)$ is:

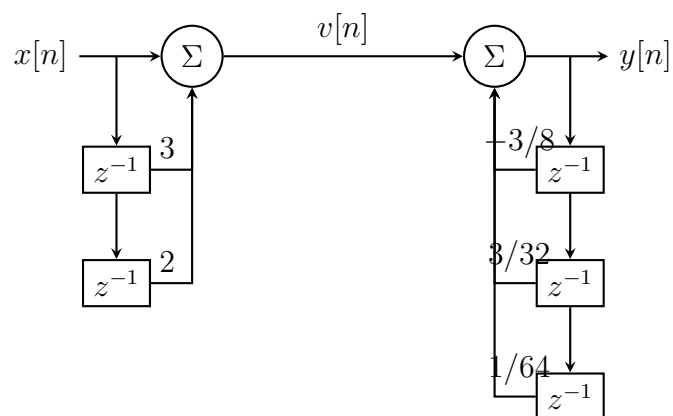
$$H(z) = \frac{1 + 3z^{-1} + 2z^{-2}}{1 + 0.375z^{-1} - 0.09375z^{-2} - 0.015625z^{-3}} \quad (73)$$

1. Direct Form-I Realization

Direct Form-I implements the transfer function as two separate cascade blocks: the all-zero (numerator) block followed by the all-pole (denominator) block. It uses separate delay lines for inputs and outputs. **Difference Equations:**

$$v[n] = x[n] + 3x[n-1] + 2x[n-2] \quad (74)$$

$$y[n] = v[n] - \frac{3}{8}y[n-1] + \frac{3}{32}y[n-2] + \frac{1}{64}y[n-3] \quad (75)$$

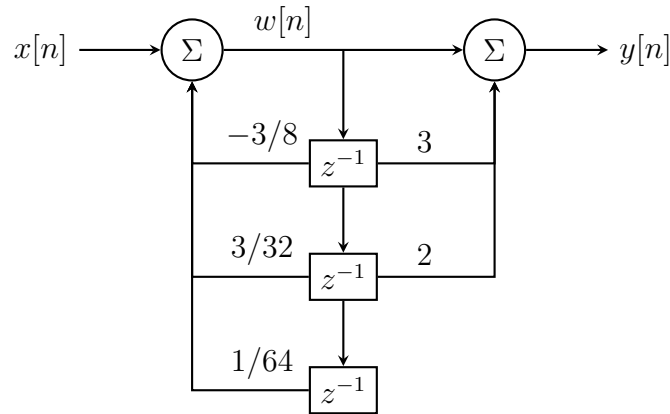


2. Direct Form-II Realization

Direct Form-II swaps the all-pole and all-zero blocks, allowing them to share a single central tapped delay line, minimizing memory requirements. We define an intermediate variable $w[n]$. **Difference Equations:**

$$w[n] = x[n] - \frac{3}{8}w[n-1] + \frac{3}{32}w[n-2] + \frac{1}{64}w[n-3] \quad (76)$$

$$y[n] = w[n] + 3w[n-1] + 2w[n-2] \quad (77)$$



3. Cascade Realization

To implement a cascade realization, we must factorize the numerator and denominator polynomials of $H(z)$ into products of first and second-order sections. **Numerator Roots:** Let $z^{-1} = v$. $2v^2 + 3v + 1 = 0 \implies (2v + 1)(v + 1) = 0 \implies v = -0.5, -1$. Numerator: $(1 + z^{-1})(1 + 2z^{-1})$.

Denominator Roots: Let $z^{-1} = v$. $1 + \frac{3}{8}v - \frac{3}{32}v^2 - \frac{1}{64}v^3 = 0$. Multiply by 64: $64 + 24v - 6v^2 - v^3 = 0$. By synthetic division, $v = 4$ is a root. $(v - 4)(-v^2 - 10v - 16) = -(v - 4)(v + 2)(v + 8)$. So the roots are $v = 4, -2, -8$. Denominator: $(1 - \frac{1}{4}z^{-1})(1 + \frac{1}{2}z^{-1})(1 + \frac{1}{8}z^{-1})$.

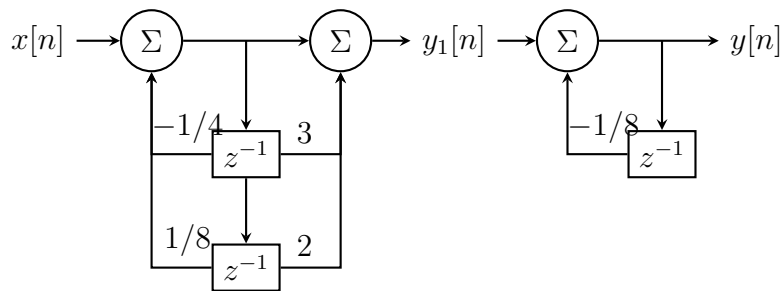
We pair the factors into a second-order section $H_1(z)$ and a first-order section $H_2(z)$:

$$H_1(z) = \frac{(1 + z^{-1})(1 + 2z^{-1})}{(1 - \frac{1}{4}z^{-1})(1 + \frac{1}{2}z^{-1})} = \frac{1 + 3z^{-1} + 2z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \quad (78)$$

$$H_2(z) = \frac{1}{1 + \frac{1}{8}z^{-1}} \quad (79)$$

Difference Equations for Cascade: Section 1: $w_1[n] = x[n] - \frac{1}{4}w_1[n - 1] + \frac{1}{8}w_1[n - 2]$ and $y_1[n] = w_1[n] + 3w_1[n - 1] + 2w_1[n - 2]$.

Section 2: $w_2[n] = y_1[n] - \frac{1}{8}w_2[n - 1]$ and $y[n] = w_2[n]$.



Question 4(b): Causal DTLTI System Analysis

Problem Statement

A causal DTLTI system is represented by the following difference equation:

$$y[n] = b_0x[n] + a_1y[n-1]$$

Show that the system is an IIR system. Determine the effect of the coefficient values on the stability of the system.

Proving it is an IIR System

To classify the system as an Infinite Impulse Response (IIR) or Finite Impulse Response (FIR) system, we must examine its impulse response sequence, $h[n]$. The impulse response is the output of the system when the input is a unit impulse, $x[n] = \delta[n]$, assuming zero initial conditions ($y[-1] = 0$).

Substituting $x[n] = \delta[n]$ and $y[n] = h[n]$ into the difference equation:

$$h[n] = b_0\delta[n] + a_1h[n-1] \quad (80)$$

Let us evaluate this recursively for $n \geq 0$:

$$\begin{aligned} n = 0 : & \quad h[0] = b_0(1) + a_1(0) = b_0 \\ n = 1 : & \quad h[1] = b_0(0) + a_1h[0] = b_0a_1 \\ n = 2 : & \quad h[2] = b_0(0) + a_1h[1] = b_0a_1^2 \\ n = 3 : & \quad h[3] = b_0(0) + a_1h[2] = b_0a_1^3 \\ & \quad \dots \\ n = k : & \quad h[k] = b_0a_1^k \end{aligned}$$

Since the system is causal, $h[n] = 0$ for $n < 0$. We can write the closed-form

expression for the impulse response using the unit step function $u[n]$:

$$h[n] = b_0(a_1)^n u[n] \quad (81)$$

As $n \rightarrow \infty$, the sequence $h[n]$ continues indefinitely (it does not strictly go to zero in finite time unless $a_1 = 0$). Because the duration of the non-zero values is infinite, this is mathematically classified as an **Infinite Impulse Response (IIR) system**.

Effect of Coefficient Values on Stability

A Discrete-Time Linear Time-Invariant (DTLTI) system is Bounded-Input Bounded-Output (BIBO) stable if and only if its impulse response is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad (82)$$

Substituting our impulse response into the stability condition:

$$\sum_{n=0}^{\infty} |b_0(a_1)^n| = |b_0| \sum_{n=0}^{\infty} |a_1|^n \quad (83)$$

This is an infinite geometric series with a common ratio of $|a_1|$. The series converges to a finite sum if and only if the common ratio is strictly less than 1.

$$\text{Sum} = \frac{|b_0|}{1 - |a_1|} \quad \text{for } |a_1| < 1 \quad (84)$$

If $|a_1| \geq 1$, the sum diverges to infinity, and the system is unstable.

Alternatively, examining the Z-domain transfer function:

$$H(z) = \frac{b_0}{1 - a_1 z^{-1}} \quad (85)$$

The system possesses a single pole at $z = a_1$. For a causal system to be BIBO

stable, all poles must lie strictly inside the unit circle of the z-plane. Therefore, we again arrive at the constraint $|a_1| < 1$.

Summary of Effects:

- b_0 : Acts purely as a scaling factor (gain). It has **no effect** on whether the system is stable or unstable.
- a_1 : Dictates stability. The system is **stable** if and only if $-1 < a_1 < 1$.

B.E. ELECTRICAL ENGINEERING - DIGITAL SIGNAL PROCESSING

PART I - DETAILED SOLUTIONS FOR QUESTION 5

Question 5

Write short notes on any two of the following (Here all four are solved completely):

- (a) Discrete-time integrators.
- (b) FIR and IIR DTLTI systems.
- (c) Representing uniform sampling as impulse modulation.
- (d) Stability of DTLTI Systems.

5(a) Discrete-Time Integrators

A discrete-time integrator is a system designed to approximate the continuous-time integration operation using discrete samples. In analog systems, an integrator is defined by the relation:

$$y(t) = \int_0^t x(\tau) d\tau \quad \xrightarrow{\text{Laplace}} \quad H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s} \quad (1)$$

To implement this digitally, we approximate the area under the curve of $x(t)$. There are three standard approximations: Forward Euler, Backward Euler, and the Trapezoidal Rule (Bilinear Transformation).

1. Backward Euler Approximation

The area is approximated by rectangles with height corresponding to the current sample $x[n]$.

$$y[n] = y[n-1] + Tx[n] \quad (2)$$

Taking the Z-transform:

$$Y(z) = z^{-1}Y(z) + TX(z) \implies H(z) = \frac{T}{1 - z^{-1}} \quad (3)$$

Mapping: $s = \frac{1-z^{-1}}{T}$.

2. Forward Euler Approximation

The area is approximated by rectangles with height corresponding to the previous sample $x[n - 1]$.

$$y[n] = y[n - 1] + Tx[n - 1] \quad (4)$$

Taking the Z-transform:

$$Y(z) = z^{-1}Y(z) + Tz^{-1}X(z) \implies H(z) = \frac{Tz^{-1}}{1 - z^{-1}} \quad (5)$$

Mapping: $s = \frac{1-z^{-1}}{Tz^{-1}}$.

3. Trapezoidal Rule (Bilinear Transformation)

This provides the most accurate approximation by calculating the area of trapezoids between samples.

$$y[n] = y[n - 1] + \frac{T}{2}(x[n] + x[n - 1]) \quad (6)$$

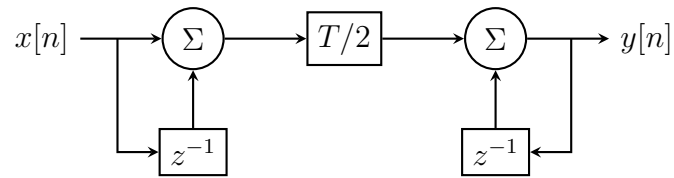
Taking the Z-transform:

$$Y(z) = z^{-1}Y(z) + \frac{T}{2}(X(z) + z^{-1}X(z)) \quad (7)$$

$$H(z) = \frac{T}{2} \frac{1 + z^{-1}}{1 - z^{-1}} \quad (8)$$

Mapping: $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$. This maps the $j\omega$ axis perfectly onto the unit circle, avoiding aliasing.

Realization of Trapezoidal Integrator:



5(b) FIR and IIR DTLTI Systems

Discrete-Time Linear Time-Invariant (DTLTI) systems are broadly classified based on the length of their impulse response $h[n]$ into Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) systems.

1. Finite Impulse Response (FIR) Systems

An FIR system is described by an impulse response that settles to zero in finite time. Mathematically, $h[n] = 0$ for $n < 0$ and $n \geq M$.

- **Difference Equation:** It only depends on present and past inputs (Non-recursive).

$$y[n] = \sum_{k=0}^{M-1} b_k x[n-k] \quad (9)$$

- **Transfer Function:**

$$H(z) = \sum_{k=0}^{M-1} b_k z^{-k} = b_0 + b_1 z^{-1} + \dots + b_{M-1} z^{-(M-1)} \quad (10)$$

- **Properties:** FIR systems have all their poles strictly at the origin ($z = 0$). Therefore, they are **inherently stable**. They can easily be designed to have exact linear phase, meaning they introduce zero phase distortion (constant group delay).

2. Infinite Impulse Response (IIR) Systems

An IIR system has an impulse response that continues indefinitely ($n \rightarrow \infty$).

- **Difference Equation:** It relies on present/past inputs AND past outputs, making it a **recursive** system with feedback.

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k] \quad (11)$$

- **Transfer Function:**

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \quad (12)$$

- **Properties:** IIR systems can have poles anywhere in the z-plane. They can become **unstable** if any pole drifts outside the unit circle ($|z| \geq 1$). They cannot achieve exact linear phase. However, they are highly efficient, achieving sharp frequency cutoffs with a much lower system order (fewer computations) compared to FIR filters.

Feature	FIR System	IIR System
Impulse Response	Finite length	Infinite length
Structure	Non-recursive (Feed-forward)	Recursive (Feedback)
Stability	Always stable	Conditionally stable
Phase Response	Strictly linear achievable	Non-linear
Memory / Order	Requires high order for sharp cut-offs	Requires lower order for sharp cutoffs

5(c) Representing Uniform Sampling as Impulse Modulation

Uniform sampling is the process of converting a continuous-time (CT) signal $x(t)$ into a discrete-time signal. Mathematically, the ideal sampling process is modelled as the multiplication of the analog signal $x(t)$ with a periodic train of ideal impulses. This is formally known as **Impulse Modulation**.

Mathematical Representation in Time Domain

Let $s(t)$ be the sampling function, defined as an infinite sum of Dirac delta functions spaced equally by the sampling period T (where $f_s = 1/T$):

$$s(t) = \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad (13)$$

The sampled signal, $x_s(t)$, is the product of the continuous signal and the impulse train:

$$x_s(t) = x(t) \cdot s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad (14)$$

Using the sifting property of the Dirac delta function ($x(t)\delta(t - nT) = x(nT)\delta(t - nT)$):

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT) \quad (15)$$

This models the sequence of samples where the weights $x(nT)$ are the discrete amplitudes, anchored to exact timestamps in the CT domain.

Frequency Domain Analysis (The Spectrum)

To understand the spectral effect, we apply the Fourier Transform. Multiplication in the time domain corresponds to convolution in the frequency domain:

$$X_s(j\omega) = \frac{1}{2\pi} [X(j\omega) * S(j\omega)] \quad (16)$$

Since $s(t)$ is periodic, its Fourier Transform is an impulse train in the frequency domain:

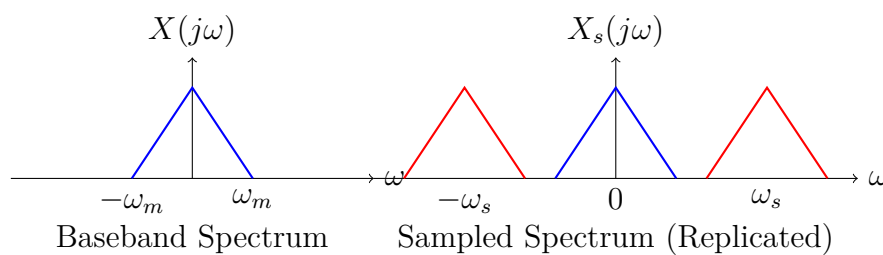
$$S(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \quad \text{where } \omega_s = \frac{2\pi}{T} \quad (17)$$

Convolving $X(j\omega)$ with $S(j\omega)$ yields the fundamental sampling equation:

$$X_s(j\omega) = \frac{1}{2\pi} \left[X(j\omega) * \left(\frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \right) \right] \quad (18)$$

$$X_s(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \quad (19)$$

Conclusion: Impulse modulation mathematically proves that sampling causes the baseband spectrum $X(j\omega)$ to be **periodically replicated** infinitely along the frequency axis at integer multiples of the sampling frequency ω_s .



5(d) Stability of DTLTI Systems

A Discrete-Time Linear Time-Invariant (DTLTI) system is declared **Bounded-Input Bounded-Output (BIBO) stable** if and only if every strictly bounded input sequence $x[n]$ produces a bounded output sequence $y[n]$. Let the input be bounded such that $|x[n]| \leq M_x < \infty$ for all n .

Time-Domain Condition

The output is related to the input via the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \quad (20)$$

We find the magnitude of the output and apply the triangle inequality ($|a+b| \leq |a|+|b|$):

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \quad (21)$$

Since $|x[n-k]| \leq M_x$:

$$|y[n]| \leq M_x \sum_{k=-\infty}^{\infty} |h[k]| \quad (22)$$

For $|y[n]|$ to be finitely bounded ($|y[n]| \leq M_y < \infty$), the summation term must be finite. Thus, the necessary and sufficient condition for BIBO stability in the time domain is that the impulse response is **absolutely summable**:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad (23)$$

Z-Domain Condition

The Z-transform of the impulse response is the system transfer function $H(z)$:

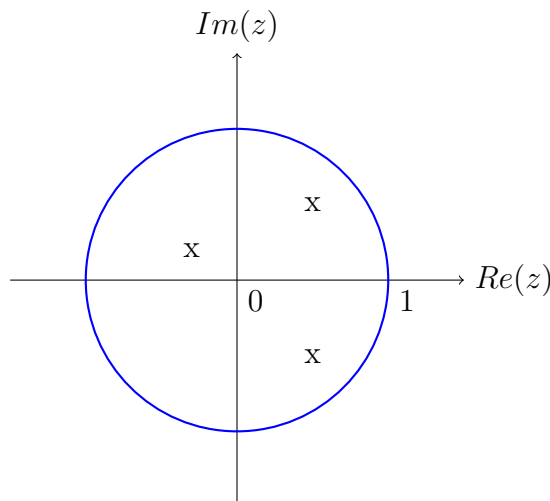
$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n} \quad (24)$$

If we evaluate this exactly on the unit circle ($z = e^{j\omega}$ where $|z| = 1$):

$$|H(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |h[n]| |e^{-j\omega n}| = \sum_{n=-\infty}^{\infty} |h[n]| \quad (25)$$

If the system is stable (absolute summability condition is met), then the series converges on the unit circle. Therefore, the fundamental Z-domain condition for stability is: **The Region of Convergence (ROC) of $H(z)$ must include the unit circle ($|z| = 1$).**

Causal Systems: For a causal DTLTI system, the sequence is right-sided, and the ROC takes the form $|z| > |p_{max}|$, where p_{max} is the pole furthest from the origin. For the ROC to include the unit circle, the outermost pole must have a magnitude less than 1. Therefore, a **causal DTLTI system is strictly stable if and only if all its poles lie strictly inside the unit circle** in the z-plane.



Stable Causal System: All poles (x) inside unit circle